

Scalability & Future Plan

Date	9 July 2026
Team ID	SWTID-2026-5447
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited to single user input	Only one prediction at a time, not suitable for bulk bank applications	High
2	Static dataset	Model accuracy may reduce over time with new credit patterns	Medium
3	Basic Security	No user login or data encryption in current Flask	Medium

Scalability Plan:

S No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports 1 user at a time on local server	Deploy on Cloud like AWS/ Heroku for 1000+ users
2	Data Storage	Using CSV file for dataset	Use SQL Database /MongoDB for large data storage
3	Performance	Prediction takes 2-3 seconds	Use optimized model + caching for faster response
4	Security	No authentication	Add user login, JWT authentication and data encryption

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase1	Core Web App with ML Model	Completed	Basic credit card approval prediction using Flask + Random Forest
Phase2	Bulk Prediction & Excel Upload	3 Months	Banks can check 1000s of application at once, saves time
Phase3	Mobile App + API Integration	6 Months	Users can check approval status from phone anywhere